Description of Additional Supplementary Files

Supplementary Data 1: All B1 β-lactamase sequences used in this study for DCA analysis.

Supplementary Data 2: All calculated DCA mutational tolerances (CDE) for 100 homologs, NDM-1 and VIM-2, listed by aligned position of each sequence.

Supplementary Data 3: All DMS fitness score data generated in this study, as the average of both replicates.

Supplementary Data 4: Data for reversion of WT residues between NDM-1 and VIM-2 based on DMS data.

Supplementary Data 5: Data for experimental reversion of WT residues from NDM-1 to VIM-2 as single or double mutants.

Supplementary Data 6: Plasmid sequence for pIDR 5.1 wtNDM-1.

Supplementary Data 7: All primers using for cloning and deep sequencing of DMS experiments.